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Fitness-for-Service Assessment - NR-21-L-0207 Measuring Point 05

Scope and Responsibilities

This document provides a Level 2 local-metal-loss evaluation for the elbow identified as NR-21-L-0207-05, located on the condensate line NR-21-L-0207 within ANLAGE-21. The assessment is triggered by wall thickness readings falling below the internal Werksnorm minimum of 2.0 mm, as documented in NR-AB-2023-04. The objective is to determine the structural integrity and remaining life of the component under design conditions of 6 bar(g) and 165 °C.

Daniel Hartmann, acting as Integrity Engineer for QSCP Corporate Asset Integrity, is responsible for the technical evaluation and corrosion rate projections. Dr. Helen Marchetti, Corporate Asset Integrity Manager, provides the final review and validation of the fitness-for-service conclusions. All calculations follow the corporate methodology defined under QSCP-CIS-001.

Inspection History and Current Condition

Periodic wall thickness monitoring recorded in RLW-PB-2023-0144 identified a minimum thickness of 1.9 mm at NR-21-L-0207-05, which initiated a nonconformity report. Subsequent high-resolution grid measurements were performed to characterize the extent of the degradation. The resulting report, RLW-PB-2023-0201, confirmed a localized minimum wall thickness of 1.7 mm over an area of approximately 40 x 60 mm.

Historical data for NR-21-L-0207 indicates a steady decline from the nominal 4.0 mm thickness. The corrosion rate has stabilized at approximately 0.108 mm/y based on the most recent survey intervals. This localized thinning is the primary subject of this assessment, as the component has now transitioned below the original design margin specified for the P235GH material.

Fitness-for-Service Evaluation

The assessment utilizes API 579-1/ASME FFS-1 to evaluate the suitability for continued operation. The basis of acceptance for the pressure-only condition is the required wall thickness of 0.22 mm, calculated in accordance with the construction edition of EN 13480. The measured minimum of 1.7 mm significantly exceeds this value, providing a substantial safety margin against internal pressure loading.

The evaluation was repeated to include supplementary loads as required by the Level 2 procedure. Under these combined loading scenarios, the component remains within acceptable stress limits. The local-metal-loss area at NR-21-L-0207-05 is verified as stable for the current design envelope of PS 6 bar(g) and TS 165 °C.

Remaining Life and Corrosion Projection

Based on the calculated corrosion rate of 0.108 mm/y, the remaining life to reach the EN 13480 required wall thickness is 13.7 years. A projection of the wall thickness indicates a value of 1.42 mm at the next scheduled re-measurement on 2026-02-12. By the planned 2028 shutdown in March 2028, the wall thickness is projected to be 1.19 mm.

These projections confirm that the elbow will maintain sufficient structural integrity through the next two operating cycles. The component is deemed acceptable for continued service at the stated design pressure and temperature, provided the monitoring and replacement schedule is strictly followed.

Recommendations and Future Actions

The elbow at NR-21-L-0207-05 shall remain in service with a mandatory re-measurement interval of 24 months to verify the projected corrosion rate. The next inspection is due by 2026-02-12. Any significant deviation from the 0.108 mm/y rate will require a revision of this fitness-for-service assessment.

Final remediation is required during the 2028 shutdown. The affected section of NR-21-L-0207 must be replaced to restore the full nominal wall thickness. Until that time, the component is fit for service under the conditions validated by Daniel Hartmann and Dr. Helen Marchetti.

Sign-off	Name	Function	Date	Signature
Prepared	Daniel Hartmann	Integrity Engineer, QSCP Corporate Asset Integrity, based at Werk Niederrhein	12.02.2024	_____
Reviewed	Dr. Helen Marchetti	Corporate Asset Integrity Manager, P.Eng.	12.02.2024	_____